gpeyt1-02 · RQD

GEO PE · Long Calculation Problems I — Topic 1: Site Characterization

Problem

A 5.0 ft (60 in.) NQ core run in a jointed limestone recovers the following intact pieces (in., measured along the core axis): 5.5, 2.0, 7.0, 3.5, 4.0, 1.5, 9.0, 3.0, 6.5, 2.5, 4.5. The rest of the run was lost as rubble. The core barrel was double-tube, drilling took 18 minutes, and the drill water return was 80%.

Most nearly, the RQD of this run and its rock-quality description are:

A. 82%, good
B. 54%, fair
C. 74%, fair
D. 61%, fair

Given equations / data

> RQD = Σ(length of intact pieces ≥ 4 in.)/(total core run length) × 100%

Worked solution

1. Pieces ≥ 4 in.: 5.5, 7.0, 4.0, 9.0, 6.5, 4.5 → Σ = 36.5 in.
2. RQD = 36.5/60 × 100 = 60.8%
3. Recovery = 49.0/60 = 82% (not RQD)
4. RQD 61% → fair (handbook RQD description table)

Final answer

61%, fair

Why the other values are wrong

- **82%, good** — reported total core recovery (all pieces) as RQD
- **54%, fair** — excluded pieces exactly 4 in. long
- **74%, fair** — divided by recovered length instead of run length

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.7.4.1 Rock Quality Designation (RQD), p. 120

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=8cX88YWUo4U (no creator values copied).

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